

Lec 41 Style GAN

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Transcript

[00:00] Hello everyone, welcome back to the lecture series on fundamentals of generative AI and large language models. In today's session, we will be discussing about style GAN which is another variation of generative adversarial network or what we call as GANs. So we will have a look into the contents for today's session. Limitations of traditional GAN and what is the motivation for style GAN. Then we will get introduced to a concept of latent space entanglement.

[00:33] Then introduction to style GAN, the mapping network, a fine transformation, synthesis network, adaptive instance normalization and then the stylebased generator as a whole. So let us start with the limitations of traditional GAN and what is the motivation for style GAN. Now we all know in traditional GAN we give a noise input to the generator and the noise is basically a random vector which is you know sampled from a normal distribution with mean zero and variance one. And this noise is basically it's a it's from the latent space which have a lot of information which is compressed or hidden inside. Okay. And when this noise goes through the generator we get an image generated image. Okay. So in this case it is a

[01:33] image of a butterfly. Okay. Now GZ means it is a generator or the noise is going through the generator G and we are getting an image. Now the limitation

of this GAN is suppose I want to change the color of the wing. Can I do that one independently without changing the pose or the posture of the butterfly? So that is the limitation. It is not possible in traditional GANs. So what is the motivation for style GAN? Suppose I have an image of a lady and then I want to change the color of the hair of this human face to blonde from brown to blonde. Now is there a way I can map that any one of these latent variables if I change I will change the I can change the color of the hair from brown to blonde but it is not there.

[02:43] Okay. So once again if I want this human face this is smiling face and this I want the smile to go off. Okay, can I have a fine control on the latent vector that I'm putting inside the generator? Can I control which latent variable I can change so that the smile goes off and it becomes a different kind of a facial expression. that is not there. Okay. And similarly, if I want to have specs, can I match it with one of the latent variables? Can I change that one and the specs will come? No, it is not there. It is not possible. So in traditional GANs, this control over the style of the face facial expressions or or the finer details it is not there. So researchers wanted finer control over the generated

[03:43] image attributes. Okay, image attributes in this case it can be the hair color, it can be the smile, it can be the glasses in the face. Okay, so those are the image attributes and this was not possible in traditional GANs. So I have taken these images from a website called thispersonexist.com. So that is entirely for you know uh explain uh explaining this concept. Now I will come to a very important concept called as latent space entanglement. Now what is this latent space entanglement? Suppose I have the image of a face of a human and once again I have the latent vector and suppose I want to change as I was telling the color the hair from brown to blonde. Now think of if I'm changing some of the latent variables

[04:51] then what happens is for changing the color first of all that mapping itself is not there that there is no specific mapping which latent variable which control which attributes just for example I'm telling that any latent variable if I change that one maybe it will change the color of the hair but at the same time it will change the shape of the face also. Okay. So it is not that independently I can change the color of the hair together with the hair color the structure of the shape also is changing. Similarly if I want here like you know the blonde color hair to change to black. I'm seeing that when this transition is happening there are some freckles that are coming. So it is very difficult to you know have

[05:48] control independently. So what the meaning is here what is the meaning of this little space entanglement uh is one latent variable affects multiple image attributes simultaneously. So if I take one of the latent variable, it may change the color of the it may change yeah it may change the color of the hair from blonde to black but at the same time there are here there was no freckles here some freckles have come that means multiple attributes are entangled and changing one latent dimen-

sion affects multiple semantic attributes. So independently we are unable to change the attributes of the image. Okay. So that is that concept is called as latent space entanglement.

[06:38] Now what is the reason for this latent space entanglement? Why this is happening? For one particular latent variable multiple attributes are associated. Why? It is told that the input latence phase must follow the probability density of the training data and that leads to unavoidable entanglements. So it is kind of a you know a textbook kind of a statement which I have put here. Now we'll understand what exactly is the meaning of uh you know probability density of the training data which leads to unavoidable entanglement. So we'll try to see what exactly it means. So here I have lot of pictures of human faces. Some are from you know some are ladies some are gens and this is a mixture. So this is my data set of human

[07:32] face. Now if we see there are if we see the short hair, the short hair is belonging to the most of the image data set in this image data set at least the short hair is belonging to the male face whereas the long hairs are belonging to the female face. Okay, the ladies. Now this is my data set. It may have other things like short hair for ladies as well as long hair for um you know gens people. Now the thing is if your model of or if our model is trained on this data set particularly where short hair is shown with the facial attributes of a male person and long hair is shown with facial attributes of a lady. So then what happens is there is an entanglement that is happening with the facial attributes and the length of the

[08:37] hair. Okay. So that means the fac multiple facial attributes that means the hair as well as the features of the face the the eyes the nose the lips those are becoming correlated. Okay and what happens is the latent space becomes curved because the generator avoids producing unlikely combination. If my data set is having equal amount of ladies and gents who are having long hair then this problem will not come. But in case our data sets are having short hair for you know gens people and long hair for ladies then the generator is unlikely to make those kind of you know uh make unlikely combinations.

[09:26] Okay. So that is the thing and this is what we call as the spa uh latent space entanglement that means with the size of the hair the facial attributes also gets correlated. Okay. Now the requirement or the motivation for getting independent control over the various attributes of human face was the main motivation for introduction to style GAN. So we will look into what is this style GAN how its architecture is different from traditional GAN. So in traditional GAN we have latent vector Z which goes to the generator. So you know you have the latent vector directly going to the generator for producing images. Whereas in style GAN you have the latent vector then you pass it through a mapping network F

[10:28] and this mapping network maps the Z latent space into an intermediate la-

latent space W and this latent space W is responsible for putting style in your image. Okay. So the main difference is the latent vector z instead of directly going to the generator in this case in the case of style GAN it goes to a mapping network F and then it goes to the generator. So in between the latent vector input and the generator there is a so this is the input and this is the generator in between there is the mapping network. Okay. And this mapping network, it maps the latent space Z into an intermediate latent space W . We will look into what is this mapping network looks like? What is its function as we go through the slides. So this is for

[11:30] style GAN and this allows different image attributes to vary independently. That means in this case in style GANs if I want to change the color of the hair I can change it independently without changing the shape of the face or if I want the smile to be reduced to be little bit reduced or increased in a facial expression that can be done independently. Okay. Now this concept of style GAN was first introduced by Tero Karas and his team.

[12:08] They are scientists from Nvidia on the paper in the paper a stylebased generator architecture for generative adversarial networks. Okay. So once again a kind of a pipeline if we can frame in our mind. So z goes to the mapping network. The mapping network gives W which is an intermediate latent space. This is responsible for giving you styles different styles and this goes to the synthesis network which is basically your generator and then an image is produced. So this is the pipeline for style GAN.

[12:46] Now we will see the mapping network and the W space in uh style GAN what exactly it is. So we have this latent vector z in case of style GAN it is z is you know sampled from a normal distribution with mean zero and variance 1. Typically z is having the dimension of 512. So this is this belongs to you know a space of 512 dimensions and this vector contains all the information in a compressed form for generating images.

[13:22] Okay. Now this goes to the latent uh sorry this goes to the mapping network F and what is this mapping network consisting of? It consists of eight layers of multi-layer perceptrons. Okay. So these are all fully connected layers and what is the function of this mapping network? It basically transforms the entangled latent space z to a more disentangled space. Okay. So the entanglement that was there in the latent space z that becomes much more disentangled in latent space w . Okay. So that is the function of the mapping network. So once you are in the latent space W you can change styles independently that means one latent variable or one direction controls the pose another direction

[14:13] controls hairstyle another direction can control smile and eyeglasses. So independent control is possible and then this goes to the generator. Now in traditional GANs the generator directly begins the image creation from z we all know. So it

goes directly it goes in traditional GAN it goes from uh the zed goes to the generator and generator generates the images in style GAN it is little different. So what happens first of all zed goes through the mapping network.

[14:46] Secondly, the style GAN it starts the generation with a fixed learned learned tensor. What is this fixed learned tensor? It is a tensor of size $4 + 4 + 512$ and 512 is coming because of the dimension of the latent space there. And this tensor is initially in uh initialized with constant values and during uh back propagation these values are updated. Okay. So this is a learned tensor. Okay. So the style GAN starts the image generation not from zed but from a learned fixed tensor or fixed learned tensor. Okay.

[15:35] Now we will have a look into the mapping network in style GAN how it looks as I was telling. So this is my latent space Z it normalize. It is normalized and then it goes to 1 2 3 4 5 6 7 8 layers of fully connected layers and we get W here and these are basically all these are multi-layer perceptrons and U the dimension is 512 and W as I was telling Z space becomes disentangled and we get all unentangled attributes untangled attributes okay and then what happens here is Till this one we have discussed. Now from W we have told that it goes to the generator. Now before it goes to the generator there is something called as the a fine transformation.

[16:26] It's a mathematical operation that is done on this latent space W. So what exactly happens here we will see. And after the fine transformation is done it goes to the synthesis network where actually the image generation happens. Okay. So we will look into what is this a fine transformation and um how the styles are getting incorporated in the uh generator network. We will uh see that one. Okay. Now as I was telling in uh traditional GANs we have uh the zed directly going to the generator and the image generation starts from there. But in style GAN it starts from a learned constant tensor. Now what exactly I mean by that one? Suppose I have a sculpture artist and this is day one where the artist have started with a clay block

[17:22] this one and then the final output that she had sculptured out is this horse. Okay. In day two, the artist once again sit with the similar block and she you know the artist creates a horse which is having different style than the day one's horse. Okay, think of this situation. So what we are seeing here same starting clay block. So both the days they have started with the same clay block day one the style of the horse sculpture of the horse is different from the style that has come out in day two. Okay. So even though we are starting from the same clay block the final outputs are different. Now we can think of this one this clay block or the starting point is the fixed constant or learned tensor. So it is fixed but

[18:31] the final output is different in style GAN. So what is the what is that part of the model which is bringing this uh difference in style? The style is injected through adaptive instance normalization which we call as ADA in. And then we have noise injection for making finer stoastic differences between images and it is not from the

initial input tensor. So the input tensor is as a starting block behaves as a starting point and then different styles are incorporated in the different layers.

[19:13] through this adaptive instance normalization which we call as ADA in and noise injection. Okay. So this is the analogy in of style D. So we can think of the learned constant tensor which is $4 + 4 + 512$ that is the starting clay block and then in the process the styles will be injected and add in have a very significant role to be played in the style injection. There are a fine transformations then add in there are convolution layers all these are a part of the style injection and then we get an output image.

[19:55] Now we will look into what is this a fine transformation in style GAN. Now a fine transformation they produce the styles. When I tell a fine transformation what exactly I mean by the term a fine transformation. Suppose I have an input X and it goes through a transformation $Y = WX + B$ where W is the trained uh weight matrix. So this is called as an a fine transformation. Okay. Now in style GAN what happens? We know the latent space now gets mapped to the latent space W . Lat space Z gets mapped to latent space W and then W goes through an aine transformation and we get Y equal to $A W + P$. And this is again one matrix and what we get the y is basically the style parameters.

[20:56] So after the a fine transformation $y = a w + b$ we get whatever y 's are coming out from there they are basically these are basically style parameters. Okay. And these style parameters get injected layer and at every layer adaptive instance normalization or adder in block at every layer. Now the fine transformation converts latent vector W into style parameter that we have already seen. Now as I was telling that the $4 \times 4 \times 512$ that is your starting clay styles are the sculpturing instructions that were that are put all throughout in the pipeline through the fine transformation and array.

[21:43] So the full architecture flow looks like this. It has two parts. Style gun has two parts. One is the mapping path. One is the synthesis part. And we have seen mapping path goes from Z to W . And synthesis part it takes care of the image generation and this produces disentangled latent space distribution. And synthesis part is the part where exactly the image generation happens and it starts from $4 + 4 + 512$ and this tensor is trainable. So it gets trained in in the process. Now how are this mapping path and the synthesis path they are connected together? How that is happening? So the connection actually happens through convolution layers.

[22:29] There are 18 convolution layers. So what exactly is happening in this 18 convolution layers. So we know this W is now going through a fine transformation and giving us Y and which are the style parameters and at every convolution layer a fine transformation produces a style parameter that are injected through ADA in so we'll look into in the next slide probably it will become more clear what exactly I want

to uh tell so you have the fixed learn tensor learned tensor which is $4 + 4 + 512$ and this is the starting point it is kept constant and then it goes to convolution layers.

[23:10] This convolution layers as I was telling there are 18 convolution layers and then these convolution layers goes through um it it produces the feature maps and then these feature maps goes through add in. Okay. Now these feature maps they contain the edges, textures, face structures, you know, uh everything. So I'll talk about what is the function of ADA in a few minutes later. Till now let us just have a look. This is the synthesis network. This is different from the mapping network. And the synthesis network starts its work from this fixed learned tensor goes to the convolution layers. There are 18 convolution layers and this convolution layers generates the feature maps which have the edges textures um structure of

[24:06] the face and everything hair pattern everything. Now what is the function of adine? I have told about it. I have you know told adinine is a very important plays a very important part in the style uh injection what exactly it does. Adin does two work. One is it normalizes the feature map. When it is normalizing a feature map it is basically erasing out all the original statistics that is there in the feature map. Okay. So it is r it is just erasing the styles that are present and then it is applying a new style using learned scales and biases from where this learned scales and biases are coming they are coming actually from the aine transformation.

[24:50] Okay. And as I was telling the convolution layers, let us put our attention here. Now I have told about what exactly is the function of ADA in here. We will look into more details of this one in the next slide. Right now let us concentrate on this convolution layers. Now there are 18 convolution layers and after every two convolution layer there is an upsampling that means the resolution improves. Okay. So let us have a look. Suppose we have convolution layer 1 and two. Then if this is upsampling this goes up samples and then three and four then five and six. And if we see the resolutions it starts from $4 + 4$ then after two convolution it goes to $8 + 8$ then $16 + 16$. And these are the low resolution part which learns the

[25:46] pose, the face shape, the code structures, the shape, the edges and the pose of the you know face everything. And then again it gets upsampled convolution 7 8 9 10 11 12 they are having the dimensions of $32 + 32$ $64 + 64$ $128 + 128$. And these middle layers learn the eyes, nose, hairstyles, everything. Okay. And then once again upsampling. So every after every two convolution layers there is an upsampling. Again we get 1314 and 1516 1718. So 1314 is 256×256 resolution. Then we have 512×512 for 15th and 16th convolution layer. And then this 17th and 18th are having 1024×1024 . So the resolution keeps improving increasing with the number of convolution layers and these higher resolution layers learn

[26:50] the pores the wrinkles the very finer details the texture of the face. So these are the features that are learned in the high resolution layer. So we have these lower

mid and high resolution layers to learn not only the coarse features but the finer textures also. Okay. Then we are coming into the adaptive instance normalization which we call as add in. Now what is this one? So as I was telling if you look into the previous slide the feature map that is generated by the convolution neural convolution layers that goes into the adine. So let my feature map be looks like look like this. And then add in does you know it normalizes the feature map and what exactly in normalization we do we take the mean and

[27:42] the you know standard deviation and we do the normalization process. So if we take the mean here so this mean is calculated like this and standard deviation is calculated for this feature map. This is the standard deviation. If there are any mistakes or any um calculation things uh mistakes are there or typos are there in these um calculations, please feel free to reach out to me or you can just um check if the calculations are um correct or wrong or you can change it by yourself also because standard deviation and mean these are very standardized uh formulas.

[28:19] So you can just look into it. Now when ADA in is normalizing the feature map it takes this you know u normalization formula $X - \mu$ divided by sigma okay which is the standard deviation and then we have zero mean and unit variance you know feature map and this is what I have done the calculations for you you can just pause and have a look after normalization What happens? The old style informations are removed. That is the main reason for doing this normalization process. Only structural information will be present.

[29:02] So this is the normalized feature map and over this one the styles will get injected. Okay. And addin first normalizes every channel independently. So every channel that is coming in the feature map this gets independently normalized by add in. Now how the style injection in adine happens. So we know that this is our latent space uh zed through the when it goes through the mapping network we come into the latent space w. The difference between zed and w is this is a entangled latent space and w is this entangled latent space and then it goes to the aine transformation which produces the style parameters. So I find transformation $y = a \cdot w + b$. And if we take you know the vector. So we have two

[29:56] components y_s and y_b these are used by ada in. Now what is this y_s and y_b these are the two components of y vector and y_s basically it is a sorry this i will not come. y_s and y_b these are the learning scaling parameters and y_b is the learned bias parameters. So there are two parameters one is the scaling parameter and one is the bias parameter. Okay. And they come out after the fine transformation. Okay. Now if I have these parameters y_s and y_b what does add in does? So we have already normalized now or this is the step of normalization if we can say this is the normalization step of the feature map. Then we are you know uh multiplying this one with y_s and y and adding this bias term y_b and this is

[30:57] done for your iat layer. Okay. So this i gives in which layer you are doing it. So this is the iat layer and layer every layer it will get changed. Okay. So add in normalizes this one. This is the normalization uh uh step. Then it gets multiplied with the learned scaling parameter and the learned bias parameter also is added here. And then we get this is the way the style is injected into the adin block. Okay. So style feature style feature maps are generated which are coming out of the adin block and we get the generated images. Okay.

[31:41] Just for giving an example if y is equal to 2 and y_b equal to 1. I have just given a numerical example. So it is $2x$ into whatever you have got in the previous normalized feature map plus one. One comes from y_b and you have this. This is basically your style feature map. This one this is your style feature map. Okay. Now so up to this we have understood how from the entangled latent space how we are coming to a disentangle latent space using the mapping network. Then we are going through the fine transformation. What is the fine transformations work? It produces the style parameters. What are the style parameters? In this case this is y_s and y_b . They are going as input to addin block. And what is adine blocks

[32:31] work? Adine first normalizes the feature maps that are coming out of the convolution layers. And by doing this process, it erases out all the statistical information that is present only keeping the structure you know intact and then style is injected through this addin block uh through this u_m style parameters. Okay. Now we'll have a whole look or uh uh look into the overall uh picture how the stylebased generator looks like. So this image I have taken from the original paper and you can look into you can go back to the paper. I urge all the participants to look into the paper lot more details you will get in the paper. So this is the traditional one where we have latent zed going into the fully connected layers

[33:23] and we get the image generation. Whereas here this is the stylebased generator and uh you see the latent uh uh entangled latent space. It goes through the mapping network. Mapping network is eight layers of fully connected uh multi-layer perceptron. Then we get this u_m W which is the uh disentangled latent space. Then these W goes to the fine transformations A for all different layers it is going. And if you see this is the synthesis network as I was telling it is starting from constant $4 + 4 + 512$.

[34:03] Then this fine transformation is giving you some style parameters which is going into ADA in and again there are convolution layers you know and uh uh then it is going through the uh process and then the thing that I want you all to look into is there is something here and there is some noise injection that is happening in these layers. Now what exactly is this one? Noise injections are happening here is u_m noise adds stochastic variation to the synthesis network. So basically very very finer details that will change that will be different from one image to the other uh image one generated image to the other generated image because we don't want exactly

the finer details. We want a stochastic variation of that one and that

[34:56] is the reason the noise gets injected. And what is B? B is a learned scaling factor. It decides how strongly the noise can affect each feature channel. So this is how the entire uh process flow works in a stylebased generator or in style GAN overall. So with this one I come to the end of my session today. uh before I conclude let us have a quick uh revision or a summary of what we have learned uh in today's discussion. Uh so we have seen that um in GANs any form of GANs fine control on the finer details or attributes of of any human face or any uh image was not possible with the traditional GA gads.

[35:48] just to have an independent control on the styles of a human face, the emotions that it depicts or other attributes associated with the face, facial expressions. That was the motivation. Can we make indep make attributes vary independently? That was the motivation for you know uh this style GAN coming into existence and people scientists discovered this architecture and uh what this architecture it have it the main difference is basically it goes it takes the latent um uh space which is an entangled one changes into a uh disentangled latent space then it gives it to the generator which generates the image And for changing the latent uh vector from the latent space which is entangled to the latent space which is

[36:45] disentangled there is a mapping network which is there which takes care it takes care of the transition from entangled latent space to the disentangle latent space. And after that one these whatever is coming out uh from the mapping network that goes through some aine transformation. These aine transformations basically give you some style parameters. So these style parameters then gets injected at various layers of the synthesis network and this synthesis network which is different from the mapping network. This starts with a learned fixed learned tensor of uh shape $4 \times 4 \times 512$ and then this is fixed and the styles get injected at every layer and the for style injection after the fine transformation whatever style parameters

[37:36] we are getting that goes and gets injected into the addin block and then addin takes care of injecting these uh styles fields into the images. What does Adine does? It first normalizes the feature map that is coming from the convolution layers. By normalizing it is actually erasing all the styles that are present there and then it is injecting or it is taking the style parameters that are coming from the fine transformation and putting them there in that feature map. Okay. And that is the way the final image gets produced. Okay. So that's the summary of the working of style GAN. With this one I conclude today's session. In my next session we will be discussing about style GAN 2 which is another variation

[38:28] of style GAN. And till we meet uh for the next session, thank you all and happy learning.